Path-integral measure for Chern–Simons theory within the stochastic quantization approach

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We discuss how the dependence of the path-integral measure on the metric affects the generating functional for the $d=3$ Chern–Simons theory. Using stochastic quantization, we show that the choice of an invariant measure preserves the topological character of the theory.

1. Introduction

The question on whether the metric dependence of the path-integral may affect properties of the generating functional Z for topological quantum field theories (TQFTs) was posed by Witten when he introduced these theories [1–3].

A discussion on this issue for the topological Yang–Mills (TYM) model in $d=4$ dimensions was given in ref. [4] using Fujikawa's prescription for defining an invariant measure [5–8]. The outcome was that the metric independence of Z is maintained in TYM theory when the metric dependence of the path-integral measure is taken into account.

In the present work we analyse this problem for another kind of TQFT, namely the Chern–Simons (CS) model in $d=3$. To this end, we will exploit a connection recently established by Baulieu [9] and Yu [10] (see also ref. [11]) between $d=3$ CS and $d=4$ TYM theories. This connection was derived using stochastic quantization without taking into account the metric dependence of the path-integral measure. As we will see, the stochastic quantization method is particularly convenient for handling the problem of metric-dependence of Z since, on the one hand, it provides a natural framework for defining Fujikawa's variables; on the other hand the introduction of stochastic time allows to end with an effective quantum action in $d=4$ dimensions for which the metric independence can be established in a simple way. Furthermore, stochastic quantization is at the root of the construction of many TQFTs [13–15].

As in any other TQFT, metric independence of the generating functional is a basic feature for the CS theory. In fact, the condition

$$\frac{\delta Z}{\delta g^{\mu\nu}}=0 \tag{1}$$

can be taken as the definition of a TQFT in the broad sense [1]: TQFTs are those theories having a vanishing energy–momentum tensor,

$$\langle T_{\mu\nu}\rangle \equiv \frac{-2}{\sqrt{g}}\frac{\delta \log Z}{\delta g^{\mu\nu}}=0\,. \tag{2}$$

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Now, the energy–momentum tensor of a Chern–Simons theory is already vanishing at the classical level since the CS action S_{CS} does not depend on the metric g^{ij}:

$$S_{CS}=\frac{k}{4\pi}\,\mathrm{tr}\int_{M^3} d^3x\, \epsilon^{ijk}(A_i\partial_j A_k+\tfrac{1}{3}A_i[A_j,A_k])\,. \quad (3)$$

Here A_i is a gauge field taking values in the Lie algebra of the gauge group G (which we will take as SU(N)), "tr" means trace in the fundamental representation and k is an integer for global reasons [16]. M^3 is some compact smooth three-manifold, $i=1, 2, 3$.

Now, quantization introduces a metric dependence in the quantum action S^q_{CS} through the gauge fixing procedure. Indeed, S^q_{CS} takes the form

$$S^q_{CS}=S_{CS}+\{Q, W\}\,, \quad (4)$$

where W, to be determined from the gauge condition, is a functional of gauge fields, ghost fields (we shall denote the collective field content as Φ) and also of the metric. In eq. (4), $\{Q, W\}$ stands for the BRST transformation of W.

One may conclude from eq. (4) that condition (2) is automatically satisfied. Indeed, from

$$\frac{\delta S^q_{CS}}{\delta g^{ij}}\,[\Phi, g^{ij}]=\left\{Q, \frac{\delta W}{\delta g^{ij}}\,[\Phi, g^{ij}]\right\} \quad (5)$$

one has

$$\left\langle\frac{\delta S^q_{CS}}{\delta g^{ij}}\right\rangle=0\,, \quad (6)$$

due to the vanishing of vacuum expectation values of BRST transformations. Now, eq. (6) implies $\delta Z_{CS}/\delta g^{ij}=0$ only if one disregards the dependence of the path-integral measure on g^{ij}. One knows however that in the measure,

$$\prod_x D\Phi(x)\,, \quad (7)$$

Π_x is non-trivial in curved space just as the volume element $d^n x$ is transformed non-trivially. Now, Fujikawa has been able to completely specify an invariant measure in curved space [5]:

$$\prod_x D\hat\Phi(x)=\prod_{x'} D\hat\Phi(x')\,, \quad (8)$$

by taking as integration variables not the original fields $\Phi=(\Phi_a)$ but appropriate tensorial densities $\hat\Phi_a$,

$$\hat\Phi_a\equiv g^{-w_a/2}\Phi_a \quad (9)$$

(w_a is a weight associated to each field Φ_a). In Fujikawa's approach the energy–momentum tensor has to be computed from Z by taking $\hat\Phi_a$ and g^{ij} as independent variables:

$$\langle\hat T_{ij}\rangle_{\hat\Phi}=-\frac{2}{\sqrt{g}}\frac{\delta\log Z_{CS}}{\delta g^{ij}}\,, \quad (10)$$

with

$$Z_{CS}=\int D\hat\Phi\exp(-\{\hat S^q_{CS}[\hat\Phi, g^{ij}]\})\,. \quad (11)$$

$D\hat\Phi$ being metric independent, $\langle\hat T_{ij}\rangle_{\hat\Phi}$ will solely pick contributions from $\hat S^q_{CS}[\hat\Phi, g_{ij}]$:

$$\langle\hat T_{ij}\rangle_{\hat\Phi}=\left\langle\frac{\delta\hat S}{\delta g^{ij}}\,[\hat\Phi, g^{ij}]\right\rangle. \quad (12)$$

It is important to notice that when one writes the quantum action $\hat S^q_{CS}$ with $\hat\Phi$ and g^{ij} as independent variables, both the classical action and the gauge fixing term give non-trivial contributions to $\langle\hat T_{ij}\rangle_{\hat\Phi}$. Indeed:

(i) The classical CS action depends on the metric when $\hat\Phi$ and g^{ij} are taken as independent variables.

(ii) Although the gauge-fixing term $\hat S_{gf}$ is still a BRST variation when written in terms of $\hat\Phi$

$$\hat S_{gf}=\{Q, \hat V[\hat\Phi, g^{ij}]\}\,, \quad (13)$$

commutation between metric and BRST variations does not necessarily hold anymore [4]. Then, one cannot conclude that $\delta\hat S_{gf}/\delta g^{ij}$ is the BRST variation of some tensor. This problem is also encountered in TYM theory. One can however find in that last case an appropriate change of field variables restoring commutativity between BRST and metric variations [4].

Then, to show that the energy–momentum tensor for a CS theory vanishes (i.e., that Z_{CS} is metric independent) one has to prove that contributions due to (i) and (ii) either cancel or vanish separately.

One may find it disgusting that having started from a metric independent action one ends with the metric entering non-trivially in the quantum action. One should note however that when quantizing within the path-integral approach, g^{ij} enters through the gauge fixing term and, what is more important, it is implicitly present in the naive path-integral measure $D\Phi$. Precisely, Fujikawa's prescription for constructing an invariant measure $D\hat{\Phi}$ has the advantage of making explicit this metric dependence at the quantum action level. This is reminiscent of anomalous gauge theories where gauge degrees of freedom, absent at the classical level, reappear after quantization [17]. In fact, the correctness of Fujikawa's approach has been confirmed in several computations of Weyl- and gravitational anomalies [6–8].

2. Stochastic quantization and Fujikawa variables

We will see now that the stochastic quantization approach provides a natural framework for the introduction of Fujikawa variables. Indeed, within this approach one introduces a stochastic time t ($t\in I\equiv[-T, T]$) and postulates a Langevin equation for fields Φ, now depending on (x, t):

$$\frac{\partial\Phi(x,t)}{\partial t}=-\frac{\delta S}{\delta\Phi(x,t)}+G(x,t)\,. \tag{14}$$

(we consider for the moment the case in which Φ is just a scalar field). Here S is the action defining the dynamics of the system and G is a gaussian noise. Denoting with $\langle\ldots\rangle_G$ stochastic expectation values, one has [18]

$$\lim_{t\to\infty}\langle\Phi_G(x_1,t)\ldots\Phi_G(x_n,t)\rangle_G$$
$$=\langle 0|\mathrm{T}\,\Phi(x_1)\ldots\Phi(x_n)|0\rangle\,, \tag{15}$$

where Φ_G denotes the solution of eq. (14) giving Φ as a functional of G and the RHS corresponds to the vacuum expectation values for the quantum field theory defined by action S.

How can one implement this schemein curved space? Since one is going to work in the presence of a metric, one needs to know the explicit dependence of the path-integral measure on it. Now, a simple way of controlling the metric dependence, which in the stochastic approach is gaussian, is to define integration variables à la Fujikawa [5]. In the present case this amounts to working with noise variables $\hat{G}$ and a stochastic measure $d\mu[\hat{G}]$ given by

$$d\mu[\hat{G}]=\prod_x d\hat{G}(x)\,\exp(-S^{(2)}[\hat{G}])\,, \tag{16}$$

with

$$S^{(2)}[\hat{G}]=\tfrac{1}{2}\int\hat{G}^{(2)}(x,t)\,d^dx\,dt \tag{17}$$

and

$$\hat{G}=g^{1/4}G\,. \tag{18}$$

In this way, the quadratic action $S^{(2)}$, when written in terms of the original variables G, takes the correct form for a gaussian action in the background of a metric $g^{\mu\nu}$ ($\mu,\nu=0, 1, 2, ..., d$; $x_0=t$, $g=\det g_{\mu\nu}$):

$$S^{(2)}[G,g^{\mu\nu}]=\tfrac{1}{2}\int G^2(x,t)\sqrt{g}\,d^dx\,dt\,. \tag{19}$$

In eqs. (17)–(19) $\hat{G}$ is the Fujikawa variable associated with the noise field $\hat{G}$. As we stated above, one has to consider $\hat{G}$ and $g^{\mu\nu}$ as independent variables. We then see that in Fujikawa's approach, the path-integral measure (17) is indeed metric independent.

Concerning stochastic expectation values, one has from (17):

$$\langle\hat{G}(x_1,t_1)\hat{G}(x_2,t_2)\rangle$$
$$=\frac{1}{\int D\mu[\hat{G}]}\int D\mu[\hat{G}]\;\hat{G}(x_1,t_1)\hat{G}(x_2,t_2)$$
$$=\delta^d(x_1-x_2)\delta(t_1-t_2)\,, \tag{20}$$

a metric independent result: all metric dependence should enter through the Langevin equation (14) which has to be written in terms of Fujikawa variables $\hat{\Phi}=g^{1/4}\Phi$ and $g^{\mu\nu}$. We then see that within the stochastic quantization approach, Fujikawa variables are the natural variables in the sense that all metric dependence is dictated by action S through the Langevin equation.

Let us apply all this machinery to the CS theory. The Langevin equation for A_i reads

$$\frac{\partial A_i(x,t)}{\partial t}=\frac{-k}{24\pi}\epsilon_{ijk}F^{jk}-D_i\Lambda+G_i(x,t)\,. \tag{21}$$

In eq. (22) G_i is a vector gaussian noise field ($i=1$, 2, 3) and the second term on the RHS induces a drift force along the gauge orbit [12]. (Λ may be an arbitrary functional of A_i). The initial condition for eq. (21) is

$$A_i(x, -T) = A_i(x) . \tag{22}$$

Since one is effectively working in a (3+1)-dimensional space, it is useful to introduce 0-components (associated with stochastic time) for the gauge fields. Following refs. [9,10] we write $A(x,t) = (-k/4\pi)A_0(x,t)$ and consider A_0 as the zeroth component of the connection A_μ defined now over a four-manifold $M = M^3 \times I$ with metric $g^{\mu\nu}$. Let A_0, till now arbitrary, satisfy the Langevin equation

$$\frac{\partial A_0}{\partial t} = \frac{-k}{4\pi} \partial_i A^i + G(x,t) , \tag{23}$$

where G is a scalar gaussian noise field. In order to write Langevin equations (21) and (23) in a covariant form, we trade the three components G_i in eq. (21) for those of a self-dual antisymmetric tensor $G_{\mu\nu}$ and write $\partial/\partial t = (k/4\pi)\partial_0$. One then has as Langevin equations:

$$F^+_{\mu\nu}[A] \equiv (F_{\mu\nu} + \tfrac{1}{2}\epsilon_{\mu\nu\alpha\beta}F^{\alpha\beta}) = G_{\mu\nu} \tag{24}$$

and

$$\partial_\mu A^\mu = G , \tag{25}$$

where we have set $e^{-1} = k/4\pi$ and redefined $e^{-1}A_\mu \to A_\mu$ so that

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu + e[A_\mu, A_\nu] . \tag{26}$$

3. The connection between CS and TYM theories in a background metric

We are now ready to find the connection between $d=4$ TYM and $d=3$ CS theories in curved space-time. We start from the stochastic generating functional for the CS theory:

$$Z^{\rm stoch}_{\rm CS} = \int \mathrm{D}\hat{G}\,\mathrm{D}\hat{G}_{ab} \exp\Big(-\tfrac{1}{2}\mathrm{tr}\int \hat{G}^2 \mathrm{d}^3x\,\mathrm{d}t - \tfrac{1}{4}\mathrm{tr}\int \hat{G}^2_{ab}\mathrm{d}^3x\,\mathrm{d}t\Big) . \tag{27}$$

G being in eq. (25) a (Lie algebra valued) scalar, $\hat{G}$ is defined as in eq. (19):

$$\hat{G} = g^{1/4} G . \tag{28}$$

Concerning $\hat{G}_{ab}$, we have introduced flat indices a, b ($a, b = 0, 1, 2, 3$) by using vielbeins $e^a_\mu(x)$,

$$G_{ab} = e^\mu_a e^\nu_b G_{\mu\nu} , \tag{29}$$

so as to use the result for scalar fields (eq. (19)) to find the Fujikawa variables associated to the noise field in eq. (24):

$$\hat{G}_{ab} = g^{1/4} G_{ab} . \tag{30}$$

One can easily return to the coordinate basis using [7]

$$\mathrm{D}\hat{G}_{ab} = \mathrm{D}g^{1/4}G_{ab} = \mathrm{D}g^{1/4}e^\mu_a e^\nu_b G_{\mu\nu} = g^{-1}\mathrm{D}g^{1/4}G_{\mu\nu} = \mathrm{D}g^{1/12}G_{\mu\nu} , \tag{31}$$

so that the invariant measure for $G_{\mu\nu}$ fields is

$$\mathrm{D}g^{1/12}G_{\mu\nu} \equiv \mathrm{D}\hat{G}_{\mu\nu} . \tag{32}$$

We can now write $Z^{\rm stoch}_{\rm CS}$ as a path integral over gauge fields A_μ. To this end, we rewrite Langevin equations (24), (25) using flat indices and Fujikawa variables:

$$F^+_{ab}[g^{-1/4}\hat{A}] = g^{-1/4}\hat{G}_{ab} , \tag{33}$$

$$\partial^a(g^{-1/4}\hat{A}_a) = g^{-1/4}\hat{G} . \tag{34}$$

From eqs. (33) and (34) we have

$$\mathrm{D}\hat{G}\,\mathrm{D}\hat{G}_{ab} = J\,\mathrm{D}\hat{A}_a \tag{35}$$

with the jacobian J given by

$$J = \det M^c_{ab} \tag{36}$$

and

$$M^c_{ab} = \begin{pmatrix} \delta\hat{G}/\delta\hat{A}_c \\ \delta\hat{G}_{ab}/\delta\hat{A}_c \end{pmatrix} . \tag{37}$$

We can now introduce ghost fields $(\hat{b}, \hat{\chi}^{ab})$ and $\hat{\psi}_c$ (with ghost number $(-1, -1)$ and 1 respectively) in order to exponentiate J

$$J = \int \mathrm{D}\hat{b}\,\mathrm{D}\hat{\chi}^{ab}\,\mathrm{D}\hat{\psi}_c \exp\Big(\mathrm{tr}\int (\hat{b}, \hat{\chi}^{ab}) M^c_{ab}\hat{\psi}_c \mathrm{d}^3x\,\mathrm{d}t\Big). \tag{38}$$

Of course, the change of variables holds for $J \neq 0$ (which corresponds to the dimension of the instanton moduli space M associated with the $F^+_{\mu\nu} = 0$ equation, $d(\mathrm{M}) = 0$, see ref. [10]). The case $J = 0$ ($d(\mathrm{M}) \neq 0$) can be treated by extending the procedure described below (see ref. [10] for details).

We see that the self-dual antisymmetric ghost $\hat{\chi}^{ab}$ and vector ghost $\hat{\psi}_a$ associated with the large (topological) symmetry of a $d=4$ TYM theory [1,13,14] naturally arises within this approach. However, as noted in ref. [9], the corresponding BRST invariance is incomplete since c-ghosts, associated with ordinary gauge symmetry, are absent. Following ref. [9] we introduce these ghosts through a Langevin equation:

$$\mathrm{D}_\mu \mathrm{D}_\nu c - \mathrm{D}_\mu \psi^\mu = \beta\,, \tag{39}$$

with β a gaussian noise. Now, inserting the identity

$$1 = \int \mathrm{D}\hat{\eta}\, \mathrm{D}\beta \exp\left[-\left(\int \mathrm{tr}\, \hat{\eta}\beta\, \mathrm{d}^3x\, \mathrm{d}t\right)\right] \tag{40}$$

in $Z_{\rm CS}^{\rm stoch}$ and changing from β to $\hat{c}$ variables via eq. (39) we finally have

$$Z_{\rm cs}^{\rm stoch} = \int \mathrm{D}\hat{A}_a \mathrm{D}\hat{b}\, \mathrm{D}\hat{\chi}^{ab} \mathrm{D}\hat{\psi}_a \mathrm{D}\hat{c}\, \mathrm{D}\hat{\eta}\, \mathrm{D}\hat{\lambda}\, \mathrm{D}\hat{\phi} \exp(-S_{\rm eff})\,, \tag{41}$$

where λ and ϕ are (Grassmann even) ghosts (with ghosts number -2 and 2 respectively) introduced in order to exponentiate the determinant associated with the change from β to $\hat{c}$ variables. The effective action reads

$$\begin{aligned} S_{\rm eff} = \mathrm{tr} \int \mathrm{d}^3x\, \mathrm{d}t \Big(& \tfrac{1}{2}\sqrt{g}\, (F^+_{ab}[\hat{A}_a g^{-1/4}])^2 \\ & + \tfrac{1}{2}\sqrt{g}\, [\partial^a(\hat{A}_a g^{-1/4})]^2 \\ & - g^{1/4}\hat{b} \frac{\delta \hat{G}}{\delta \hat{A}_p} g^{-1/4}\hat{\psi}_p + g^{1/4}\hat{\chi}^{ab} \frac{\delta \hat{G}_{ab}}{\delta \hat{A}_p} g^{-1/4}\hat{\psi}_p \\ & + g^{1/4}\hat{\lambda} \frac{\delta \beta}{\delta \hat{A}_p} g^{-1/4}\hat{\psi}_p + g^{1/4}\hat{\lambda} \frac{\delta \beta}{\delta \hat{c}} g^{-1/4}\phi \\ & + g^{1/4}\hat{\eta}[\mathrm{D}_a \mathrm{D}_a(\hat{c} g^{-1/4}) - \mathrm{D}_a(\hat{\psi}_a g^{-1/4})] \Big)\,. \end{aligned} \tag{42}$$

It is easy to write $S_{\rm eff}$ in the coordinate basis:

$$\begin{aligned} S_{\rm eff} = \mathrm{tr} \int \sqrt{g}\, \mathrm{d}^3x\, \mathrm{d}t \{ & \tfrac{1}{2}(F^+_{\mu\nu})^2 + \tfrac{1}{2}(\partial^\mu A^\mu)^2 \\ & - \eta \mathrm{D}_\mu \mathrm{D}_\mu c - \chi^{\mu\nu} \mathrm{D}_{[\mu,} \psi_{\nu]} - b \partial_\mu \psi^\mu - \eta \mathrm{D}_\mu \psi^\mu \\ & + \lambda(\mathrm{D}_\mu \mathrm{D}^\mu \phi + [\psi^\mu, \psi^\mu + \mathrm{D}_\mu c] + \mathrm{D}_\mu[\psi^\mu, c])\} \end{aligned} \tag{43}$$

Since, as explained above, physical quantities have to be calculated using (hatted) Fujikawa variables, we list below the connection between hatted and unhatted fields in the coordinate basis:

$$\begin{aligned} \hat{\Phi} &= (\hat{A}_\mu, \hat{\chi}^{\mu\nu}, \hat{\psi}^\mu, \hat{b}, \hat{c}, \hat{\eta}, \hat{\lambda}, \hat{\phi}) \\ &= (g^{1/8}A_\mu, g^{5/12}\chi^{\mu\nu}, g^{1/8}\psi_\mu, g^{1/4}b, \\ &\quad g^{1/4}c, g^{1/4}\eta, g^{1/4}\lambda, g^{1/4}\phi)\,. \end{aligned} \tag{44}$$

The effective action (43) has a full topological invariance:

$$\begin{aligned} &\{Q, A_\mu\} = \psi_\mu, \quad \{Q, \psi_\mu\} = 0\,, \\ &\{Q, b\} = G, \quad \{Q, G\} = 0\,, \\ &\{Q, \chi^{\mu\nu}\} = G_{\mu\nu}, \quad \{Q, G_{\mu\nu}\} = 0\,, \\ &\{Q, c\} = \phi, \quad \{Q, \phi\} = 0\,, \\ &\{Q, \lambda\} = \eta, \quad \{Q, \eta\} = 0\,, \end{aligned} \tag{45}$$

and can be written in the Q-exact form:

$$S_{\rm eff} = \{Q, V_{\rm CS}[\Phi]\}\,, \tag{46}$$

with

$$\begin{aligned} V_{\rm CS} = \mathrm{tr}\{ & \chi^{\mu\nu} F^+_{\mu\nu} - \tfrac{1}{4}\chi^{\mu\nu} G_{\mu\nu} + b(\partial_\mu A^\mu + \tfrac{1}{2}G) \\ & - \lambda(\mathrm{D}_\mu \psi^\mu + \mathrm{D}_\mu \mathrm{D}^\mu c)\}\,. \end{aligned} \tag{47}$$

(Note that in (45)–(47) we have reintroduced the noise fields G and $G_{\mu\nu}$.) One recognizes in (46) the gauge fixing action arising from quantization of the second Chern–Simons class [1]. That is, $S_{\rm eff}$ corresponds to the quantum action of a $d=4$ TYM theory. Note that in order to compare $S_{\rm eff}$ given by eqs. (46), (47) with that usually found in the literature [1,4,13,14], one has to change variables ψ_μ and ϕ as follows:

$$\psi_\mu \to \psi_\mu + \mathrm{D}_\mu c\,, \tag{48}$$

$$\phi \to \phi - \tfrac{1}{2}[c, c]\,. \tag{49}$$

We will however work with $S_{\rm eff}$ as in eqs. (46), (47) since the corresponding BRST transformations (45), when written in terms of Fujikawa variables, commute with metric variations. (If one proceeds to the change of variables (48), (49) then the identity $\delta\{Q, \hat{\Phi}\}/\delta g^{\mu\nu} = \{Q, \delta\hat{\Phi}/\delta g^{\mu\nu}\}$ is no more valid). In this form, one can easily prove the metric independence of $Z_{\rm CS}^{\rm stoch}$. Indeed, writing

$$Z^{\rm stoch}_{\rm CS} = \int {\rm D}\Phi \exp(-\{Q, V[\Phi, g^{\mu\nu}]\}) \tag{50}$$

and differentiating with respect to $g^{\mu\nu}$ one has

$$\frac{\delta Z^{\rm stoch}_{\rm CS}}{\delta g^{\mu\nu}} = \int {\rm D}\Phi \left\{Q, \frac{\delta V}{\delta g^{\mu\nu}}\right\} \exp(-\{Q, V[\Phi, g^{\mu\nu}]\}) \\ =0\,. \tag{51}$$

(The last equality was established using BRST invariance of the path-integral measure [1]).

From eq. (51) one concludes that the CS partition function, within the stochastic quantization approach, is metric independent. The proof basically relies on the fact that in this approach the effective action can be written as the BRST transformation of some functional $V_{\rm CS}$, even when the dependence on the metric is taken into account.

Note that as immediate consequence of this fact, independence of $Z^{\rm stoch}_{\rm CS}$ on k follows. This can be easily seen by factorizing the coupling constant $e=4\pi/k$ (see eq. (26)) so that $S_{\rm eff}=(1/e^2)\{Q, V_{\rm CS}\}$. One can then use Witten's result [1] on the independence of the generating functional for TYM theory on e^2 since, as we stated above, $S_{\rm eff}=S_{\rm TYM}$. Concerning the evaluation of topological invariants using the connection between CS and TYM theories, let us note that the modifications of the stochastic quantization procedure need whenever $d({\rm M})\neq 0$ make the analysis more involved [10]. However the conclusions of ref. [10] about the connection between topological invariants of TYM theory and those arising in $d=3$ CS theory should continue to hold when the dependence of the path-integral measure on the metric is taken into account using Fujikawa variables.

Let us conclude this work by summarizing what we learnt from the study of metric dependence of the path-integral measure for TQFTs. Using Fujikawa variables to define an invariant measure, we proved in ref. [4] that the TYM generating functional is indeed metric independent. In fact, in Fujikawa's approach, the dependence on the metric is transferred to the quantum action $S_{\rm q}$ by an appropriate choice of integration variables. Since in terms of these new variables $S_{\rm q}$ can be still written as the BRST transform of some functional, $S_{\rm q}=\{Q, V\}$, with $[Q, \delta/\delta g^{\mu\nu}]=0$, one finally has $\delta Z/\delta g^{\mu\nu}=0$. This property can be proved analogously for any other TQFT arising as TYM from quantization of trivial classical actions.

Concerning CS theory in $d=3$, the classical action is not trivial and the quantum action cannot be written as the BRST transform of some functional. One knows however, from the works of Baulieu [9] and Yu [10], that within the stochastic quantization approach, the introduction of stochastic time allows to connect the generating functional for CS theory in $d=3$ with that of an effective action in $d=3+1$ for which the quantum action can be written as $S_{\rm eff}=\{Q, V_{\rm CS}\}$. In fact, one finds that $V_{\rm CS}=V_{\rm TYM}$. In the present work we have proved this connection taking into account the metric dependence of the path-integral measure. Once this is done, metric independence of the stochastic generating functional for CS theory in $d=3$ follows.

Finally, it is interesting to note that in the stochastic quantization framework, the Fujikawa measure in curved space stems as a natural one. One should then expect that the evaluation of Wilson loops and other expectation values for CS theory will be simpler in this approach.

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